

Submitted by:
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PROJECT REPORT

Healthcare Management System

Hospital Administrative Digitization Initiative

Project Structure & File Manifest

File	Description
db.js	Contains the SQL Server configuration (sa user credentials, 127.0.0.1 server connection, and connection pool logic).
server.js	The main entry point for the backend. Handles all HTTP routing, API requests, and database query execution.
App.js	The core frontend component. Manages the UI layout, form handling for creating admission records, and state management for viewing logs.
package.json	Lists all dependencies, including express, mssql, and cors, required for project execution.

Files

[db.js](#)

```
const sql = require('mssql');

const config = {
  user: "sa",
  password: "Admin@Database786",
  server: "127.0.0.1",
  database: "Healthcare Management System",
```

```

options: {
  instanceName: "SQLEXPRESS",
  trustServerCertificate: true,
  enableArithAbort: true,
  encrypt: false
},
pool: {
  max: 10,
  min: 0,
  idleTimeoutMillis: 30000
}
};

const poolPromise = new sql.ConnectionPool(config)
  .connect()
  .then(pool => {
    console.log("✅ DB Optimized Connection Pool Ready!");
    return pool;
  })
  .catch(err => {
    console.log("❌ DB Connection Error Details: ", err.message);
  });

module.exports = { sql, poolPromise };

```

server.js

```

const express = require("express");
const sql = require("mssql");
const cors = require("cors");
const { poolPromise } = require("./db");

const app = express();

app.use(cors());
app.use(express.json());

// =====
// BASE ROUTE
// =====
app.get("/", (req, res) => {
  res.send("Healthcare API Running Smoothly!");
});

// =====
// PATIENTS API
// =====

// Get Patients
app.get("/patients", async (req, res) => {
  try {

```

```

const pool = await poolPromise;

const result = await pool.request().query(`
  SELECT TOP 500 *
  FROM Patient.Patients
  ORDER BY PatientID DESC
`);

res.json(result.recordset || []);
} catch (err) {
  console.error("Patients Fetch Error:", err.message);
  res.status(500).send(err.message);
}
});

// Add Patient
app.post("/patients", async (req, res) => {
  try {
    const { name, age, gender, bloodType } = req.body;

    const pool = await poolPromise;

    await pool.request().query(`
      INSERT INTO Patient.Patients
      (Name, Age, Gender, BloodType)
      VALUES
      ('${name}', ${age}, '${gender}', '${bloodType}')
    `);

    res.send("Patient Inserted Successfully");
  } catch (err) {
    console.error(err.message);
    res.status(500).send(err.message);
  }
});

// Delete Patient
app.delete("/patients/:id", async (req, res) => {
  try {
    const { id } = req.params;

    const pool = await poolPromise;

    await pool.request().query(`
      DELETE FROM Patient.Patients
      WHERE PatientID = ${id}
    `);

    res.send("Patient deleted successfully");
  } catch (err) {
    console.error("Delete Error:", err.message);
  }
});

```

```

        res.status(500).send("Database error: Could not delete record.");
    }
});

// =====
// DROPDOWNS API
// =====

app.get("/dropdowns", async (req, res) => {
    try {
        const pool = await poolPromise;

        const doctors = await pool.request().query(`
            SELECT TOP 500 *
            FROM Medical.Doctors
        `);

        const hospitals = await pool.request().query(`
            SELECT TOP 500 *
            FROM Hospital.Hospitals
        `);

        const insurance = await pool.request().query(`
            SELECT TOP 500 *
            FROM Billing.InsuranceProviders
        `);

        const conditions = await pool.request().query(`
            SELECT TOP 500 *
            FROM Medical.Conditions
        `);

        res.json({
            doctors: doctors.recordset || [],
            hospitals: hospitals.recordset || [],
            insurance: insurance.recordset || [],
            conditions: conditions.recordset || []
        });

    } catch (err) {
        console.error("Dropdown Error:", err.message);
        res.status(500).send(err.message);
    }
});

// =====
// GET ADMISSIONS
// =====

app.get("/admissions", async (req, res) => {
    try {

```

```

const pool = await poolPromise;

const result = await pool.request().query(`
  SELECT TOP 500
    A.AdmissionID,
    A.PatientID,
    P.Name AS PatientName,
    A.RoomNumber,
    A.BillingAmount
  FROM Hospital.Admissions A
  INNER JOIN Patient.Patients P
    ON A.PatientID = P.PatientID
  ORDER BY A.AdmissionID DESC
`);

res.json(result.recordset || []);

} catch (err) {
  console.error("Admissions Fetch Error:", err.message);
  res.status(500).send(err.message);
}
});

// =====
// ADD ADMISSION
// =====

app.post("/admissions", async (req, res) => {
  try {
    const {
      PatientID,
      DoctorID,
      HospitalID,
      InsuranceID,
      ConditionID,
      AdmissionDate,
      DischargeDate,
      RoomNumber,
      BillingAmount,
      Medication,
      TestResults
    } = req.body;

    const pool = await poolPromise;

    await pool.request()
      .input("pID", sql.Int, PatientID || null)
      .input("dID", sql.Int, DoctorID || null)
      .input("hID", sql.Int, HospitalID || null)
      .input("iID", sql.Int, InsuranceID || null)

```

```

.input("clD", sql.Int, ConditionID || null)
.input("ad", sql.Date, AdmissionDate || null)
.input("dd", sql.Date, DischargeDate || null)
.input("rn", sql.Int, RoomNumber || null)
.input("ba", sql.Decimal(18, 2), BillingAmount || 0)
.input("med", sql.NVarChar, Medication || "")
.input("tr", sql.NVarChar, TestResults || "")
.query(`
    INSERT INTO Hospital.Admissions
    (
        PatientID,
        DoctorID,
        HospitalID,
        InsuranceID,
        ConditionID,
        AdmissionDate,
        DischargeDate,
        RoomNumber,
        BillingAmount,
        Medication,
        TestResults
    )
    VALUES
    (
        @pID,
        @dID,
        @hID,
        @iID,
        @clD,
        @ad,
        @dd,
        @rn,
        @ba,
        @med,
        @tr
    )
`);

```

```

res.send("Admission Record Saved Successfully");

```

```

} catch (err) {
    console.error("Admission Insert Error:", err.message);
    res.status(500).send(err.message);
}
});

```

```

// =====
// SERVER START
// =====

```

```

app.listen(5000, () => {

```

```
    console.log("🚀 High-Speed Optimized Backend Running on Port 5000");
  });
```

App.js

```
import { useEffect, useState } from "react";
```

```
function App() {
```

```
  const API = "http://localhost:5000";
```

```
  // Navigation Control
```

```
  const [activeTab, setActiveTab] = useState("dashboard");
```

```
  const [openFAQ, setOpenFAQ] = useState(null);
```

```
  // SUB-TABS CONTROL: Set defaults back to show "add" form first on screen load to utilize handlers
```

```
  const [patientSubTab, setPatientSubTab] = useState("add");
```

```
  const [admissionSubTab, setAdmissionSubTab] = useState("add");
```

```
  // Live Database States
```

```
  const [patients, setPatients] = useState([]);
```

```
  const [admissions, setAdmissions] = useState([]);
```

```
  const [searchQuery, setSearchQuery] = useState("");
```

```
  // LOCAL STORAGE TRACKERS
```

```
  const [extraPatients, setExtraPatients] = useState(() => {
    return parseInt(localStorage.getItem("extraPatients") || "0", 10);
  });
```

```
  const [extraAdmissions, setExtraAdmissions] = useState(() => {
    return parseInt(localStorage.getItem("extraAdmissions") || "0", 10);
  });
```

```
  // Dashboard Metrics Counter State
```

```
  const [stats, setStats] = useState({ totalPatients: 0, totalDoctors: 0, totalAdmissions: 0 });
```

```
  // Dropdowns Structure
```

```
  const [dropdowns, setDropdowns] = useState({
    doctors: [], hospitals: [], insurance: [], conditions: []
  });
```

```
  // Patient Registration Form State
```

```
const [patientForm, setPatientForm] = useState({
  name: "", age: "", gender: "", bloodType: ""
});
```

// Admission Logging Form State

```
const [admissionForm, setAdmissionForm] = useState({
  PatientID: "",
  DoctorID: "",
  HospitalID: "",
  InsuranceID: "",
  ConditionID: "",
  AdmissionDate: "",
  DischargeDate: "",
  RoomNumber: "",
  BillingAmount: "",
  Medication: "",
  TestResults: ""
});
```

// Hospital Operational Desk Form State

```
const [ticketForm, setTicketForm] = useState({ name: "", type: "Ward Resource Reallocation",
message: "" });
```

// Live Sync Data Load Hook

```
useEffect(() => {
  const loadAllData = async () => {
    try {
      let pLength = 0;
      let dLength = 0;
      let aLength = 0;

      const resP = await fetch(`${API}/patients`);
      if (resP.ok) {
        const dataP = await resP.json();
        setPatients(dataP || []);
        pLength = dataP ? dataP.length : 0;
      }

      const resD = await fetch(`${API}/dropdowns`);
      let dataD = { doctors: [], hospitals: [], insurance: [], conditions: [] };
      if (resD.ok) {
```

```

    dataD = await resD.json();
    setDropdowns(dataD);
    dLength = dataD.doctors ? dataD.doctors.length : 0;
  }

  const resA = await fetch(`${API}/admissions`);
  if (resA.ok) {
    const dataA = await resA.json();
    setAdmissions(dataA || []);
    aLength = dataA ? dataA.length : 0;
  }

  setStats({
    totalPatients: pLength + extraPatients,
    totalDoctors: dLength,
    totalAdmissions: aLength + extraAdmissions
  });

} catch (err) {
  console.error("Data synchronization failed:", err);
}
};

loadAllData();
}, [extraPatients, extraAdmissions]);

// Filter client data safely based on Search Query
const filteredPatients = patients.filter(p => {
  const nameMatch = p.Name ? p.Name.toLowerCase().includes(searchQuery.toLowerCase())
: false;
  const bloodMatch = p.BloodType ?
p.BloodType.toLowerCase().includes(searchQuery.toLowerCase()) : false;
  return nameMatch || bloodMatch;
});

// Form Value Trackers
const handlePatientChange = (e) => {
  setPatientForm({ ...patientForm, [e.target.name]: e.target.value });
};

const handleAdmissionChange = (e) => {

```

```

    setAdmissionForm({ ...admissionForm, [e.target.name]: e.target.value });
};

// Operations Form Tracker
const handleTicketChange = (e) => {
  setTicketForm({ ...ticketForm, [e.target.name]: e.target.value });
};

// ACTION 1: Submit Patient Profile

const deletePatient = async (id) => {
  if (window.confirm("Are you sure you want to delete this patient?")) {
    try {
      const res = await fetch(`${API}/patients/${id}`, { method: "DELETE" });

      if (res.ok) {
        // 1. LocalStorage se count kam karein
        const currentCount = parseInt(localStorage.getItem("extraPatients") || "0", 10);
        if (currentCount > 0) {
          const newCount = currentCount - 1;
          setExtraPatients(newCount); // State update
          localStorage.setItem("extraPatients", newCount.toString()); // Storage update
        }

        alert("✅ Patient successfully deleted and count updated!");
        window.location.reload();
      } else {
        // Purane "alert" ko hata kar ye line daal dein
        const errorData = await res.text();
        alert(`❌ Server Error: ${errorData}`);
      }
    } catch (err) {
      console.error(err);
      alert("❌ Connection error.");
    }
  }
};

const handlePatientSubmit = async (e) => {

  e.preventDefault();
  if (!patientForm.name || !patientForm.age || !patientForm.gender || !patientForm.bloodType) {

```

```

    alert("⚠ Error: Please make sure all patient inputs are completely filled!");
    return;
}
try {
    const dataToSend = {
        name: patientForm.name,
        age: parseInt(patientForm.age, 10),
        gender: patientForm.gender,
        bloodType: patientForm.bloodType
    };
    const res = await fetch(`${API}/patients`, {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify(dataToSend)
    });
    if (res.ok) {
        const nextCount = extraPatients + 1;
        setExtraPatients(nextCount);
        localStorage.setItem("extraPatients", nextCount.toString());

        alert("🎉 Patient Record Successfully Saved!");
        setPatientForm({ name: "", age: "", gender: "", bloodType: "" });
        setPatientSubTab("view");
        window.location.reload();
    } else {
        const errText = await res.text();
        alert("❌ Error: ${errText}");
    }
} catch (err) {
    console.error(err);
    alert("❌ Link Failure: Cannot reach hospital local server.");
}
};

// ACTION 2: Submit Admission Record
// React component mein handleAdmissionSubmit function
const handleAdmissionSubmit = async (e) => {
    e.preventDefault();

    // Mapping state to backend expected keys
    const dataToSend = {

```

```

PatientID: parseInt(admissionForm.patientId) || 0,
DoctorID: parseInt(admissionForm.doctorId) || 0,
HospitalID: parseInt(admissionForm.hospitalId) || 0,
InsuranceID: parseInt(admissionForm.insuranceId) || 0,
ConditionID: parseInt(admissionForm.conditionId) || 0,
AdmissionDate: admissionForm.admissionDate,
DischargeDate: admissionForm.dischargeDate,
RoomNumber: parseInt(admissionForm.roomNumber) || 0, // Ensure integer
BillingAmount: parseFloat(admissionForm.billingAmount) || 0.0, // Ensure decimal
Medication: admissionForm.medication || "",
TestResults: admissionForm.testResults || ""
};

try {
  const response = await fetch("http://localhost:5000/admissions", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify(dataToSend),
  });

  if (!response.ok) {
    const errorText = await response.text();
    throw new Error(errorText);
  }
  alert("Record saved successfully!");
} catch (err) {
  alert("Error saving record: " + err.message);
}

// ACTION 3: Submit Operational Request
const handleTicketSubmit = (e) => {
  e.preventDefault();
  alert(`Thank you Officer ${ticketForm.name}! Your emergency coordination alert regarding "${ticketForm.type}" has been routed to the Clinical Response Command.`);
  setTicketForm({ name: "", type: "Ward Resource Reallocation", message: "" });
};

return (
  <div style={{ display: "flex", height: "100vh", width: "100vw", backgroundColor: "#f8f9fa",
fontFamily: "'Segoe UI', Tahoma, sans-serif", overflow: "hidden" }}>

```

```

    {/ *  LEFT PANEL SIDEBAR NAVIGATION */}
    <div style={{ width: "260px", backgroundColor: "#0f172a", color: "#f8fafc", display: "flex",
    flexDirection: "column", padding: "25px 0", boxShadow: "4px 0 15px rgba(0,0,0,0.05)",
    flexShrink: 0 }}>
      <div style={{ padding: "0 24px 25px 24px", borderBottom: "1px solid #1e293b", display:
    "flex", alignItems: "center", gap: "12px" }}>
        <span style={{ fontSize: "30px" }}><img alt="Building icon" data-bbox="441 198 461 213"/></span>
        <div>
          <h2 style={{ fontSize: "18px", fontWeight: "700", margin: 0, color: "#38bdf8",
    letterSpacing: "0.5px" }}>CarePortal</h2>
          <span style={{ fontSize: "11px", color: "#64748b", fontWeight: "600" }}>HOSPITAL
    MANAGEMENT</span>
        </div>
      </div>

      <div style={{ display: "flex", flexDirection: "column", gap: "8px", padding: "25px 12px" }}>
        <button onClick={() => setActiveTab("dashboard")} style={{ padding: "12px 18px",
    borderRadius: "10px", border: "none", cursor: "pointer", textAlign: "left", fontSize: "14px",
    fontWeight: "600", transition: "all 0.2s", backgroundColor: activeTab === "dashboard" ?
    "#38bdf8" : "transparent", color: activeTab === "dashboard" ? "#0f172a" : "#94a3b8" }}>
          <img alt="Bar chart icon" data-bbox="176 456 196 471"/> Overview Dashboard
        </button>
        <button onClick={() => setActiveTab("patients")} style={{ padding: "12px 18px",
    borderRadius: "10px", border: "none", cursor: "pointer", textAlign: "left", fontSize: "14px",
    fontWeight: "600", transition: "all 0.2s", backgroundColor: activeTab === "patients" ? "#38bdf8" :
    "transparent", color: activeTab === "patients" ? "#0f172a" : "#94a3b8" }}>
          <img alt="Two people icon" data-bbox="176 566 196 581"/> Patients Registry
        </button>
        <button onClick={() => setActiveTab("admissions")} style={{ padding: "12px 18px",
    borderRadius: "10px", border: "none", cursor: "pointer", textAlign: "left", fontSize: "14px",
    fontWeight: "600", transition: "all 0.2s", backgroundColor: activeTab === "admissions" ?
    "#38bdf8" : "transparent", color: activeTab === "admissions" ? "#0f172a" : "#94a3b8" }}>
          <img alt="Calendar icon" data-bbox="176 671 196 686"/> Admission Logs
        </button>
        <button onClick={() => setActiveTab("help")} style={{ padding: "12px 18px", borderRadius:
    "10px", border: "none", cursor: "pointer", textAlign: "left", fontSize: "14px", fontWeight: "600",
    transition: "all 0.2s", backgroundColor: activeTab === "help" ? "#38bdf8" : "transparent", color:
    activeTab === "help" ? "#0f172a" : "#94a3b8" }}>
          ? Helpdesk Support
        </button>
      </div>
    </div>

    {/ *  MAIN WORKSPACE LAYOUT */}

```

```
<div style={{ flex: 1, padding: "40px", display: "flex", flexDirection: "column", height: "100%",
overflowY: activeTab === "help" || (activeTab === "patients" && patientSubTab === "add") ||
(activeTab === "admissions" && admissionSubTab === "add") ? "auto" : "hidden" }}>
```

```
  { /* TAB 1: OVERVIEW DASHBOARD */ }
```

```
  { activeTab === "dashboard" && (
```

```
    <div style={{ overflowY: "auto", flex: 1 }}>
```

```
      <div style={{ background: "linear-gradient(135deg, #1e3a8a 0%, #0f172a 100%)",
padding: "40px", borderRadius: "20px", color: "white", marginBottom: "40px", boxShadow: "0
10px 25px rgba(30, 58, 138, 0.12)" }}>
```

```
        <h1 style={{ margin: 0, fontSize: "32px", fontWeight: "700", letterSpacing: "-0.5px"
}}>Welcome to CarePortal Core</h1>
```

```
        <p style={{ margin: "10px 0 0 0", color: "#93c5fd", fontSize: "15px", fontWeight: "500",
lineHeight: "1.5" }}>
```

```
          Enterprise Healthcare Management Dashboard. Live analytical view of medical
records and hospital statistics.
```

```
        </p>
```

```
      </div>
```

```
    <div style={{ display: "grid", gridTemplateColumns: "repeat(auto-fit, minmax(240px,
1fr))", gap: "30px" }}>
```

```
      <div style={{ backgroundColor: "#ffffff", padding: "30px", borderRadius: "20px",
boxShadow: "0 4px 20px rgba(0,0,0,0.01)", borderLeft: "6px solid #3b82f6" }}>
```

```
        <div style={{ display: "flex", justifyContent: "space-between", alignItems: "center" }}>
```

```
          <p style={{ margin: 0, color: "#64748b", fontWeight: "700", fontSize: "12px",
letterSpacing: "1px" }}>TOTAL PATIENTS</p>
```

```
          <span style={{ fontSize: "22px", backgroundColor: "#eff6ff", padding: "10px",
borderRadius: "12px" }}><img alt="Patients icon" data-bbox="315 585 335 605"/></span>
```

```
        </div>
```

```
        <h2 style={{ margin: "15px 0 0 0", color: "#0f172a", fontSize: "38px", fontWeight: "800"
}}>{stats.totalPatients}</h2>
```

```
      </div>
```

```
    <div style={{ backgroundColor: "#ffffff", padding: "30px", borderRadius: "20px",
boxShadow: "0 4px 20px rgba(0,0,0,0.01)", borderLeft: "6px solid #10b981" }}>
```

```
      <div style={{ display: "flex", justifyContent: "space-between", alignItems: "center" }}>
```

```
        <p style={{ margin: 0, color: "#64748b", fontWeight: "700", fontSize: "12px",
letterSpacing: "1px" }}>TOTAL DOCTORS</p>
```

```
        <span style={{ fontSize: "22px", backgroundColor: "#f0fdf4", padding: "10px",
borderRadius: "12px" }}><img alt="Doctors icon" data-bbox="315 805 335 825"/></span>
```

```
      </div>
```

```
        <h2 style={{ margin: "15px 0 0 0", color: "#0f172a", fontSize: "38px", fontWeight: "800"
}}>{stats.totalDoctors}</h2>
```

```
    </div>
```

```

    <div style={{ backgroundColor: "#ffffff", padding: "30px", borderRadius: "20px",
    boxShadow: "0 4px 20px rgba(0,0,0,0.01)", borderLeft: "6px solid #ef4444" }}>
      <div style={{ display: "flex", justifyContent: "space-between", alignItems: "center" }}>
        <p style={{ margin: 0, color: "#64748b", fontWeight: "700", fontSize: "12px",
        letterSpacing: "1px" }}>ACTIVE ADMISSIONS</p>
        <span style={{ fontSize: "22px", backgroundColor: "#fef2f2", padding: "10px",
        borderRadius: "12px" }}>🍷</span>
      </div>
      <h2 style={{ margin: "15px 0 0 0", color: "#0f172a", fontSize: "38px", fontWeight: "800"
      }}>{stats.totalAdmissions}</h2>
    </div>
  </div>
</div>
)}

```

```

{ /* TAB 2: PATIENTS REGISTRY */

```

```

{activeTab === "patients" && (

```

```

  <div style={{ display: "flex", flexDirection: "column", height: "100%", overflow: "hidden" }}>

```

```

    { /* SUB NAV BAR FOR PATIENTS */

```

```

      <div style={{ display: "flex", justifyContent: "space-between", alignItems: "center",
      marginBottom: "25px", flexShrink: 0 }}>

```

```

        <div style={{ display: "flex", gap: "10px", backgroundColor: "#e2e8f0", padding: "4px",
        borderRadius: "10px" }}>

```

```

          <button onClick={() => setPatientSubTab("add")} style={{ padding: "8px 16px", border:
          "none", borderRadius: "8px", cursor: "pointer", fontWeight: "600", fontSize: "13px", transition: "all
          0.2s", backgroundColor: patientSubTab === "add" ? "white" : "transparent", color: patientSubTab
          === "add" ? "#0f172a" : "#64748b", boxShadow: patientSubTab === "add" ? "0 2px 5px
          rgba(0,0,0,0.05)" : "none" }}>

```

```

            + Register New Patient

```

```

          </button>

```

```

          <button onClick={() => setPatientSubTab("view")} style={{ padding: "8px 16px",
          border: "none", borderRadius: "8px", cursor: "pointer", fontWeight: "600", fontSize: "13px",
          transition: "all 0.2s", backgroundColor: patientSubTab === "view" ? "white" : "transparent", color:
          patientSubTab === "view" ? "#0f172a" : "#64748b", boxShadow: patientSubTab === "view" ? "0
          2px 5px rgba(0,0,0,0.05)" : "none" }}>

```

```

            📋 View Active Records

```

```

          </button>

```

```

        </div>

```

```

      {patientSubTab === "view" && (

```

```

        <input

```

```

          type="text"

```

```

placeholder="🔍 Search Patients..."
value={searchQuery}
onChange={(e) => setSearchQuery(e.target.value)}
style={{ width: "300px", padding: "10px 16px", borderRadius: "10px", border: "1px
solid #cbd5e1", fontSize: "13px" }}
/>
})
</div>

{/* SUB-TAB VIEW: REGISTER CLEAN FORM */}
{patientSubTab === "add" && (
  <div style={{ backgroundColor: "#ffffff", padding: "35px", borderRadius: "20px",
boxShadow: "0 4px 25px rgba(0,0,0,0.02)", maxWidth: "600px", margin: "0 auto", width: "100%"
}}>
    <h3 style={{ margin: "0 0 25px 0", color: "#0f172a", fontSize: "18px", fontWeight:
"700", borderBottom: "2px solid #f1f5f9", paddingBottom: "12px" }}>👤 Register New
Patient</h3>
    <form onSubmit={handlePatientSubmit} style={{ display: "flex", flexDirection:
"column", gap: "18px" }}>
      <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
        <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Full
Name</label>
        <input name="name" required value={patientForm.name} placeholder="e.g. Ali
Khan" onChange={handlePatientChange} style={{ padding: "12px 15px", borderRadius: "10px",
border: "1px solid #cbd5e1" }} />
      </div>
      <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
        <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Age</label>
        <input name="age" type="number" required value={patientForm.age}
placeholder="e.g. 28" onChange={handlePatientChange} style={{ padding: "12px 15px",
borderRadius: "10px", border: "1px solid #cbd5e1" }} />
      </div>
      <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
        <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569"
}}>Gender</label>
        <select name="gender" required value={patientForm.gender}
onChange={handlePatientChange} style={{ padding: "12px 15px", borderRadius: "10px", border:
"1px solid #cbd5e1", backgroundColor: "white" }}>
          <option value="">Select Gender</option>
          <option value="Male">Male</option>
          <option value="Female">Female</option>
        </select>
      </div>
      <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>

```

```

        <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Blood
Type</label>
        <input name="bloodType" required value={patientForm.bloodType}
placeholder="e.g. O+" onChange={handlePatientChange} style={{ padding: "12px 15px",
borderRadius: "10px", border: "1px solid #cbd5e1" }} />
    </div>
    <button type="submit" style={{ padding: "14px", backgroundColor: "#3b82f6", color:
"white", border: "none", borderRadius: "10px", cursor: "pointer", fontSize: "14px", fontWeight:
"700", marginTop: "10px" }}>
        Save Patient Profile
    </button>
</form>
</div>
))

```

```

{ /* SUB-TAB VIEW: DATA TABLE */ }
{patientSubTab === "view" && (
    <div style={{ backgroundColor: "#ffffff", padding: "35px", borderRadius: "20px",
boxShadow: "0 4px 25px rgba(0,0,0,0.02)", display: "flex", flexDirection: "column", flex: 1,
minHeight: 0 }}>
        <h3 style={{ margin: "0 0 20px 0", color: "#0f172a", fontSize: "18px", fontWeight:
"700", borderBottom: "2px solid #f1f5f9", paddingBottom: "12px", flexShrink: 0 }}>📋 Active
Patient Records</h3>
        <div style={{ flex: 1, overflowY: "auto", overflowX: "auto" }}>
            <table style={{ width: "100%", borderCollapse: "collapse", fontSize: "14px" }}>
                <thead style={{ position: "sticky", top: 0, backgroundColor: "#ffffff", zIndex: 1 }}>
                    <tr style={{ backgroundColor: "#f8f9fa", color: "#64748b", textAlign: "left",
borderBottom: "2px solid #edf2f7" }}>
                        <th style={{ padding: "14px" }}>Patient Name</th>
                        <th style={{ padding: "14px" }}>Age</th>
                        <th style={{ padding: "14px" }}>Gender</th>
                        <th style={{ padding: "14px" }}>Blood Group</th>
                        <th style={{ padding: "14px" }}>Action</th>
                    </tr>
                </thead>
                <tbody>
                    {filteredPatients.length === 0 ? (
                        <tr><td colspan="4" style={{ padding: "25px", textAlign: "center", color:
"#94a3b8" }}>No patient records found.</td></tr>
                    ) : (
                        filteredPatients.slice(0, 100).map((p, i) => (
                            <tr key={i} style={{ borderBottom: "1px solid #f1f5f9" }}>

```

```

        <td style={{ padding: "14px", fontWeight: "600", color: "#1e293b"
}}>{p.Name}</td>
        <td style={{ padding: "14px", color: "#475569" }}>{p.Age}</td>
        <td style={{ padding: "14px", color: "#475569" }}>{p.Gender}</td>
        <td style={{ padding: "14px", color: "#ef4444", fontWeight: "700"
}}>{p.BloodType}</td>
        <td style={{ padding: "14px" }}>
        <button
        onClick={() => deletePatient(p.PatientID)}
        style={{ cursor: "pointer", color: "white", backgroundColor: "#ef4444", border: "none", padding:
"5px 10px", borderRadius: "5px" }}
        >
        Delete
        </button>
</td>
    </tr>
    ))
    }}
</tbody>
</table>
</div>
</div>
    )}
</div>
    )}

```

```

{ /* TAB 3: ADMISSIONS LOGS */}
{activeTab === "admissions" && (
    <div style={{ display: "flex", flexDirection: "column", height: "100%", overflow: "auto" }}>

        { /* SUB NAV BAR FOR ADMISSIONS */}
        <div style={{ display: "flex", marginBottom: "25px", flexShrink: 0 }}>
            <div style={{ display: "flex", gap: "10px", backgroundColor: "#e2e8f0", padding: "4px",
borderRadius: "10px" }}>
                <button onClick={() => setAdmissionSubTab("add")} style={{ padding: "8px 16px",
border: "none", borderRadius: "8px", cursor: "pointer", fontWeight: "600", fontSize: "13px",
transition: "all 0.2s", backgroundColor: admissionSubTab === "add" ? "white" : "transparent",
color: admissionSubTab === "add" ? "#0f172a" : "#64748b", boxShadow: admissionSubTab ===
"add" ? "0 2px 5px rgba(0,0,0,0.05)" : "none" }}>
                    + Create Admission Record
                </button>
                <button onClick={() => setAdmissionSubTab("view")} style={{ padding: "8px 16px",
border: "none", borderRadius: "8px", cursor: "pointer", fontWeight: "600", fontSize: "13px",

```

```
transition: "all 0.2s", backgroundColor: admissionSubTab === "view" ? "white" : "transparent",
color: admissionSubTab === "view" ? "#0f172a" : "#64748b", boxShadow: admissionSubTab
=== "view" ? "0 2px 5px rgba(0,0,0,0.05)" : "none" }>
```

```
📄 View Admission Logs
```

```
</button>
```

```
</div>
```

```
</div>
```

```
{/* SUB-TAB VIEW: CENTRALIZED ENHANCED FORM */}
```

```
{admissionSubTab === "add" && (
```

```
<div style={{ backgroundColor: "#ffffff", padding: "35px", borderRadius: "20px",
boxShadow: "0 4px 25px rgba(0,0,0,0.02)", maxWidth: "700px", margin: "0 auto", width: "100%"
}}>
```

```
<h3 style={{ margin: "0 0 25px 0", color: "#0f172a", fontSize: "18px", fontWeight:
"700", borderBottom: "2px solid #f1f5f9", paddingBottom: "12px" }}>🏠 Create Admission
Record</h3>
```

```
<form onSubmit={handleAdmissionSubmit} style={{ display: "flex", flexDirection:
"column", gap: "18px" }}>
```

```
<div style={{ display: "grid", gridTemplateColumns: "1fr 1fr", gap: "20px" }}>
```

```
<div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
```

```
<label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Select
Patient</label>
```

```
<select name="patientId" value={admissionForm.patientId}
onChange={handleAdmissionChange} style={{ padding: "12px", borderRadius: "10px", border:
"1px solid #cbd5e1", backgroundColor: "white" }}>
```

```
<option value="">-- Choose Patient --</option>
```

```
{patients.slice(0, 200).map((p) => <option key={p.PatientID}
value={p.PatientID}>{p.Name}</option>)}
```

```
</select>
```

```
</div>
```

```
<div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
```

```
<label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Assign
Doctor</label>
```

```
<select name="doctorId" value={admissionForm.doctorId}
onChange={handleAdmissionChange} style={{ padding: "12px", borderRadius: "10px", border:
"1px solid #cbd5e1", backgroundColor: "white" }}>
```

```
<option value="">-- Choose Doctor --</option>
```

```
{dropdowns.doctors.map((d) => <option key={d.DoctorID}
value={d.DoctorID}>{d.DoctorName}</option>)}
```

```
</select>
```

```
</div>
```

```
</div>
```

```
<div style={{ gridTemplateColumns: "1fr 1fr", gap: "20px", display: "grid" }}>
```

```

    <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
      <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Hospital
Location</label>
      <select name="hospitalId" value={admissionForm.hospitalId}
onChange={handleAdmissionChange} style={{ padding: "12px", borderRadius: "10px", border:
"1px solid #cbd5e1", backgroundColor: "white" }}>
        <option value="">-- Choose Hospital --</option>
        {dropdowns.hospitals.map((h) => <option key={h.HospitalID}
value={h.HospitalID}>{h.HospitalName}</option>)}
      </select>
    </div>
    <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
      <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Insurance
Provider</label>
      <select name="insuranceId" value={admissionForm.insuranceId}
onChange={handleAdmissionChange} style={{ padding: "12px", borderRadius: "10px", border:
"1px solid #cbd5e1", backgroundColor: "white" }}>
        <option value="">-- Choose Insurance --</option>
        {dropdowns.insurance.map((i) => <option key={i.InsuranceID}
value={i.InsuranceID}>{i.ProviderName}</option>)}
      </select>
    </div>
  </div>

  <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
    <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Diagnosed
Condition</label>
    <select name="conditionId" value={admissionForm.conditionId}
onChange={handleAdmissionChange} style={{ padding: "12px", borderRadius: "10px", border:
"1px solid #cbd5e1", backgroundColor: "white" }}>
      <option value="">-- Choose Condition --</option>
      {dropdowns.conditions.map((c) => <option key={c.ConditionID}
value={c.ConditionID}>{c.ConditionName}</option>)}
    </select>
  </div>

  <div style={{ display: "grid", gridTemplateColumns: "1fr 1fr", gap: "20px" }}>
    <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
      <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Admission
Date</label>
      <input type="date" name="admissionDate"
value={admissionForm.admissionDate} onChange={handleAdmissionChange} style={{ padding:
"11px", borderRadius: "10px", border: "1px solid #cbd5e1" }} />
    </div>
  </div>

```

```

        <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
            <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Discharge
Date</label>
            <input type="date" name="dischargeDate" value={admissionForm.dischargeDate}
onChange={handleAdmissionChange} style={{ padding: "11px", borderRadius: "10px", border:
"1px solid #cbd5e1" }} />
        </div>
    </div>

    <div style={{ display: "grid", gridTemplateColumns: "1fr 1fr", gap: "20px" }}>
        <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
            <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Room
Number</label>
            <input type="number" name="roomNumber" value={admissionForm.roomNumber}
placeholder="e.g. 104" onChange={handleAdmissionChange} style={{ padding: "12px",
borderRadius: "10px", border: "1px solid #cbd5e1" }} />
        </div>
        <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
            <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Billing
Amount ($)</label>
            <input type="number" name="billingAmount"
value={admissionForm.billingAmount} placeholder="e.g. 2500"
onChange={handleAdmissionChange} style={{ padding: "12px", borderRadius: "10px", border:
"1px solid #cbd5e1" }} />
        </div>
    </div>

    <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
        <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Prescribed
Medication</label>
        <input name="medication" value={admissionForm.medication} placeholder="e.g.
Paracetamol" onChange={handleAdmissionChange} style={{ padding: "12px", borderRadius:
"10px", border: "1px solid #cbd5e1" }} />
    </div>

    <div style={{ display: "flex", flexDirection: "column", gap: "6px" }}>
        <label style={{ fontSize: "13px", fontWeight: "600", color: "#475569" }}>Test Results
Summary</label>
        <input name="testResults" value={admissionForm.testResults} placeholder="e.g.
Normal Stable" onChange={handleAdmissionChange} style={{ padding: "12px", borderRadius:
"10px", border: "1px solid #cbd5e1" }} />
    </div>

```

```
        <button type="submit" style={{ padding: "14px", backgroundColor: "#ef4444", color:
"white", border: "none", borderRadius: "10px", marginTop: "10px", cursor: "pointer", fontWeight:
"700", fontSize: "14px" }}>Save Admission Record</button>
```

```
    </form>
```

```
  </div>
```

```
}}
```

```
    { /* SUB-TAB VIEW: DATA TABLE */ }
```

```
{admissionSubTab === "view" && (
```

```
  <div style={{ backgroundColor: "#ffffff", padding: "35px", borderRadius: "20px", boxShadow: "0
4px 25px rgba(0,0,0,0.02)", display: "flex", flexDirection: "column", flex: 1, minHeight: 0 }}>
```

```
    <h3 style={{ margin: "0 0 20px 0", color: "#0f172a", fontSize: "18px", fontWeight: "700",
borderBottom: "2px solid #f1f5f9", paddingBottom: "12px", flexShrink: 0 }}>  Admissions
Logs</h3>
```

```
    <div style={{ flex: 1, overflowY: "auto", overflowX: "auto" }}>
```

```
      <table style={{ width: "100%", borderCollapse: "collapse", fontSize: "13px" }}>
```

```
        <thead style={{ position: "sticky", top: 0, backgroundColor: "#ffffff", zIndex: 1 }}>
```

```
          <tr style={{ backgroundColor: "#f8f9fa", color: "#64748b", textAlign: "left", borderBottom:
"2px solid #edf2f7" }}>
```

```
            <th style={{ padding: "12px" }}>AdmissionID</th>
```

```
            <th style={{ padding: "12px" }}>Patient ID</th>
```

```
            <th style={{ padding: "12px" }}>Patient Name</th>
```

```
            <th style={{ padding: "12px" }}>Room</th>
```

```
            <th style={{ padding: "12px" }}>Billing</th>
```

```
          </tr>
```

```
        </thead>
```

```
        <tbody>
```

```
          {admissions.map((a, i) => (
```

```
            <tr key={i} style={{ borderBottom: "1px solid #f1f5f9" }}>
```

```
              <td style={{ padding: "12px" }}>{a.AdmissionID}</td>
```

```
              <td style={{ padding: "12px" }}>{a.PatientID}</td>
```

```
              <td style={{ padding: "12px", fontWeight: "600" }}>{a.PatientName}</td>
```

```
              <td style={{ padding: "12px" }}>{a.RoomNumber}</td>
```

```
              <td style={{ padding: "12px" }}>${a.BillingAmount}</td>
```

```
            </tr>
```

```
          )))
```

```
        </tbody>
```

```
      </table>
```

```
    </div>
```

```
  </div>
```

```
}}
```

```
  </div>
```

```
}}
```

```
{/* TAB 4: HELP & SUPPORT */}
{activeTab === "help" && (
  <div style={{ padding: "20px" }}>
    <div style={{ marginBottom: "35px" }}>
      <h2 style={{ color: "#0f172a", fontSize: "28px", fontWeight: "700" }}>Clinical Support &
Coordination Desk</h2>
      <p style={{ color: "#64748b", fontSize: "15px" }}>
        Access immediate hospital emergency protocols, manage floor distributions, or log
operational requests directly to the main administration panel.
      </p>
    </div>
```

```
    {/* Top 3 Info Cards */}
    <div style={{ display: "grid", gridTemplateColumns: "repeat(3, 1fr)", gap: "20px",
marginBottom: "40px" }}>
      <div style={{ background: "white", padding: "20px", borderRadius: "15px", borderTop:
"4px solid #ef4444", boxShadow: "0 2px 10px rgba(0,0,0,0.05)" }}>
        <h4>Emergency Triage Hot-Line</h4>
        <p style={{ fontSize: "20px", fontWeight: "bold", color: "#ef4444" }}>+92 (51)
111-222-333</p>
        <small>Main ER Trauma Ward Routing Desk (24/7)</small>
      </div>
      <div style={{ background: "white", padding: "20px", borderRadius: "15px", borderTop:
"4px solid #3b82f6", boxShadow: "0 2px 10px rgba(0,0,0,0.05)" }}>
        <h4>Chief Medical Superintendent Office</h4>
        <p style={{ color: "#3b82f6", fontWeight: "600"
}}>superintendent@careportalpro.com</p>
        <small>Clinical advisory response matrix under 15 mins</small>
      </div>
      <div style={{ background: "white", padding: "20px", borderRadius: "15px", borderTop:
"4px solid #10b981", boxShadow: "0 2px 10px rgba(0,0,0,0.05)" }}>
        <h4>Medical Logistics & Inventory</h4>
        <p style={{ color: "#10b981", fontWeight: "bold" }}>Active Operations</p>
        <small>Emergency pharmacy and ward gear coordination</small>
      </div>
    </div>
```

```
    {/* Bottom Section: FAQ and Form */}
    <div style={{ display: "grid", gridTemplateColumns: "1fr 1fr", gap: "30px" }}>
```

```
    {/* INTERACTIVE FAQ SECTION */}
```

```
<div style={{ background: "white", padding: "25px", borderRadius: "15px" }}>
  <h3 style={{ marginBottom: "20px" }}>Clinical Operations & Protocols FAQ</h3>
```

```
  <div onClick={() => setOpenFAQ(openFAQ === 1 ? null : 1)} style={{ padding: "15px",
border: "1px solid #e2e8f0", borderRadius: "8px", marginBottom: "10px", cursor: "pointer",
backgroundColor: openFAQ === 1 ? "#f8f9fc" : "transparent" }}>
```

```
    ▶ How do I update missing patient profiles or incorrect Blood Groups?
```

```
    {openFAQ === 1 && <p style={{ marginTop: "10px", color: "#475569", fontSize:
"14px" }}>Navigate to the "Patients Registry" tab and click on the patient's profile to edit
details.</p>}
```

```
  </div>
```

```
  <div onClick={() => setOpenFAQ(openFAQ === 2 ? null : 2)} style={{ padding: "15px",
border: "1px solid #e2e8f0", borderRadius: "8px", cursor: "pointer", backgroundColor: openFAQ
=== 2 ? "#f8f9fc" : "transparent" }}>
```

```
    ▶ What is the process for ward bed allocation changes?
```

```
    {openFAQ === 2 && <p style={{ marginTop: "10px", color: "#475569", fontSize:
"14px" }}>Submit a request using the "Clinical Operations Alert" form on the right selecting
"Ward Resource Reallocation".</p>}
```

```
  </div>
```

```
</div>
```

```
<form onSubmit={handleTicketSubmit} style={{ background: "white", padding: "25px",
borderRadius: "15px" }}>
```

```
  <h3 style={{ marginBottom: "20px" }}>Submit Clinical Operations Alert</h3>
```

```
  <label style={{ fontSize: "13px", fontWeight: "600" }}>On-Duty Medical Staff / Admin
Officer</label>
```

```
  <input name="name" placeholder="e.g. Dr. Ayesha" required
value={ticketForm.name} onChange={handleTicketChange} style={{ width: "100%", padding:
"10px", marginTop: "5px", marginBottom: "15px", borderRadius: "8px", border: "1px solid
#cbd5e1" }} />
```

```
  <label style={{ fontSize: "13px", fontWeight: "600" }}>Incident Coordination
Category</label>
```

```
  <select name="type" value={ticketForm.type} onChange={handleTicketChange}
style={{ width: "100%", padding: "10px", marginTop: "5px", marginBottom: "15px", borderRadius:
"8px", border: "1px solid #cbd5e1" }}>
```

```
    <option>Ward Resource Reallocation</option>
```

```
    <option>Emergency Patient Transfer</option>
```

```
  </select>
```

```
  <label style={{ fontSize: "13px", fontWeight: "600" }}>Request / Operational Incident
Description</label>
```

```

    <textarea name="message" required value={ticketForm.message}
onChange={handleTicketChange} style={{ width: "100%", padding: "10px", marginTop: "5px",
marginBottom: "15px", borderRadius: "8px", border: "1px solid #cbd5e1", height: "80px" }} />

    <button type="submit" style={{ width: "100%", padding: "12px", background:
"#0f172a", color: "white", border: "none", borderRadius: "8px", cursor: "pointer" }}>Submit
Alert</button>
    </form>
  </div>
</div>
)}
</div>
</div>
);
}

```

export default App;

Package.json

```

{
  "name": "backend",
  "version": "1.0.0",
  "description": "",
  "main": "server.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "dependencies": {
    "cors": "^2.8.6",
    "express": "^5.2.1",
    "mssql": "^12.5.5",
    "nodemon": "^3.1.14"
  }
}

```

Frontend

